IAM's Law

The Thermodynamic Cost of Classical Existence

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Abstract

We present and derive a candidate thermodynamic law governing the cost of classical existence. Every irreversible transition from quantum potential to classical actuality — every decoherence event on a timelike worldline — pays a thermodynamic cost of $k_B T_H \ln 2$ per bit to the nearest encoding surface. This cost is physically real, irreversible, and cosmologically consequential. We call this IAM's Law.

The law has two expressions at two scales. Locally, it provides the first complete physical interpretation of the virial theorem, established by Clausius in 1870 [**?**]: the kinetic energy $\langle K \rangle$ of any $1/r$ bound system is the Landauer cost of the system's transition from unbound potential to bound actuality. The virial theorem $2\langle K \rangle + \langle V \rangle = 0$ is the mechanical face of this identity. The thermodynamic completion $\langle K \rangle = Q = T\,\Delta S = E_{\text{Landauer}}$ is its other face. The $1/2$ partition is the unique ratio at which mechanical and thermodynamic equilibrium are simultaneously satisfied for a $1/r$ potential.

Globally, the accumulated Landauer cost of 13.8 billion years of decoherence events constitutes a missing term in the Jacobson–Cai-Kim entropy functional. Adding it produces the activation function $\mathcal{E}(a) = \exp(1 - 1/a)$, a sector-dependent gravitational coupling $\mu(a) < 1$ for matter and $\Sigma(a) = 1$ for photons, and a coupling constant $\beta_m = \Omega_m/2$ derived from the virial theorem with zero free parameters. This cosmological model is one derived implementation of the broader law; the law itself operates locally through the virial theorem and globally through the horizon entropy completion.

The law is applied across 37 orders of magnitude with one equation and zero free parameters. The cosmological prediction $\beta_m = \Omega_m/2 = 0.1575$ is consistent with 17 independent converged MCMC chains against the full Planck 2018 likelihood at 0.2σ [**?**].

Keywords: IAM's Law; thermodynamic cost of actuality; virial theorem; Landauer's principle; gravitational decoherence; Jacobson–Cai-Kim entropy; horizon thermodynamics; dark energy; σ_8 tension; H_0 tension

1 The Law in Plain Language

Before the mathematics, a plain statement of what this paper claims.

The universe contains two kinds of things: things that exist classically — with definite positions, definite states, definite identities — and things that exist only as quantum potential — superpositions of possibilities that have not yet resolved into definite outcomes. The transition from the second kind to the first kind is called decoherence. It happens constantly, everywhere: every atom that binds, every molecule that forms, every star that collapses, every galaxy that virializes.

Here is the claim: that transition is not free. Every time something crosses from quantum potential to classical actuality, the universe charges a thermodynamic price. That price is paid irreversibly to the boundary of the observable universe — the cosmic horizon — at a rate of $k_B T_H \ln 2$ per bit of classical information produced. Once paid, the cost cannot be recovered. Classical existence is continuously purchased, one decoherence event at a time.

This is not a metaphor. It is a physical claim with a derivation, a mechanism, and measured consequences.

Locally, the virial theorem $2\langle K\rangle + \langle V\rangle = 0$ has governed the energy of bound physical systems since Clausius stated it in 1870 [?]. For any system bound by a $1/r$ potential — a hydrogen atom, a solar system, a galaxy — the kinetic energy at equilibrium is exactly half the magnitude of the potential energy. This is one of the most broadly verified results in physics, confirmed from the electron orbital to the cosmic web. But the virial theorem has always been stated as a mechanical result. It does not explain what $\langle K\rangle$ physically is, nor where the other half of $|\langle V\rangle|$ goes when a system virializes. That question went unanswered for 154 years.

IAM's Law answers it. The other half of $|\langle V\rangle|$ is the heat Q released irreversibly at the virialization boundary — the thermodynamic cost of the system's transition from unbound potential to bound actuality. By Landauer's principle [?], this cost equals the kinetic energy retained in the bound state: $Q = \langle K\rangle = E_{\text{Landauer}}$. The virial theorem is the mechanical face of this identity. The 1/2 partition is not arbitrary. It is the unique ratio at which mechanical and thermodynamic equilibrium are simultaneously satisfied for a $1/r$ potential. One coin, two sides.

Globally, Jacobson showed in 1995 that the Einstein field equations follow from thermodynamics applied to local causal horizons [?]. Cai and Kim extended this to the FRW apparent horizon in 2005, recovering the Friedmann equations [?]. Both derivations assume the entropy functional of the horizon contains only geometric terms. IAM's Law says that assumption is incomplete: the Landauer cost of every irreversible decoherence event in the bulk must also be encoded on the horizon. This is the term that Einstein's GR lacked. It is what Jacobson was one entropy term short of. Adding it completes the derivation that Jacobson began.

The virial theorem then serves as the balance sheet at the cosmological scale: it tells you exactly how much energy crosses the actualization boundary at each virialization event, giving $\beta_m = \Omega_m/2$ with no free parameters. Seventeen converged MCMC chains against the full Planck 2018 likelihood are consistent with this prediction at 0.2σ.

The law is the same at both scales. The cost is the same. The balance sheet is the virial theorem. The only difference is the scale at which the ledger is read.

2 Statement of the Law

Law 1 (IAM's Law)**.** *Every irreversible transition from quantum potential to classical actuality — every decoherence event on a timelike worldline — pays a thermodynamic cost of $k_B T_H \ln 2$ per bit of classical information produced, encoded permanently on the nearest available encoding surface at the local horizon temperature T_H:*

$$\Delta E_{\text{act}} = k_B T_H \ln 2 \quad \textit{per bit.} \tag{1}$$

The transition is irreversible. The cost cannot be recovered. Potential moves toward actual; never the reverse.

Three consequences follow immediately:

Corollary 1 (The thermodynamic arrow of time)**.** *Each payment of the actualization cost is irreversible. The accumulated ledger of all such payments, tracked by $\mathcal{E}(a)$, is monotonically increasing. The thermodynamic arrow of time is a structural property of IAM's Law, derived quantitatively in Section 11.*

3 The Hierarchy

IAM's Law generates a complete hierarchy at two scales.

IAM's Law [Section 2] — the root principle: every crossing pays $k_B T_H \ln 2$ per bit.

Local expression [Section 5]: IAM's Law applied to $1/r$ bound systems.

- *The virial theorem derived*: the equilibrium condition $2\langle K\rangle + \langle V\rangle = 0$ follows from Euler's theorem for homogeneous functions of degree -1, not assumed.
- *Thermodynamic completion*: $\langle K\rangle = Q = T\,\Delta S = E_{\text{Landauer}}$. The physical meaning of K, identified for the first time.
- *The $1/2$ partition*: the unique ratio at which mechanical and thermodynamic equilibrium are simultaneously satisfied for $1/r$. Verified across 37 orders of magnitude.

Global expression [Section 6]: IAM's Law integrated across cosmological time.

- *The missing entropy term*: accumulated Landauer cost of 13.8 billion years of decoherence events, encoded on the cosmic horizon — the term Jacobson and Cai-Kim lacked.
- *The activation function*: $\mathcal{E}(a) = \exp(1 - 1/a)$, derived from horizon thermodynamics.
- *The virial theorem as balance sheet*: $\beta_m = \Omega_m/2$, connecting the local $1/2$ partition to the cosmological coupling constant. Zero free parameters.
- *The dual-sector split*: $\mu(a) < 1$ for matter on timelike worldlines; $\Sigma(a) = 1$ for photons on null geodesics. Geometric necessity, not assumption.

The bridge [Section 8]: the critical exponent $n = 5/2$, derived independently from both directions. Their convergence is a nontrivial consistency check that the same law operates at both scales.

4 The Quantum Boundary of Reality

IAM's Law operates at the boundary between quantum and classical. To state it precisely, we must state what that boundary is.

4.1 The Decoherence Event

Consider a quantum system $\mathcal{S}$ coupled to an environment $\mathcal{E}$. An initial product state evolves under the total Hamiltonian into an entangled state [?]:

$$|\psi_S\rangle \otimes |E_0\rangle \;\longrightarrow\; \sum_i c_i\,|s_i\rangle \otimes |E_i\rangle, \tag{2}$$

where $\{|s_i\rangle\}$ are the pointer states selected by the system-environment interaction. When the environmental states become approximately orthogonal ($\langle E_i|E_j\rangle \approx \delta_{ij}$), the reduced density matrix of $\mathcal{S}$ becomes diagonal:

$$\rho_S = \mathrm{Tr}_E\,|\Psi\rangle\langle\Psi| \;\longrightarrow\; \sum_i |c_i|^2\,|s_i\rangle\langle s_i|. \tag{3}$$

Coherence is lost. The system has crossed from quantum superposition to classical actuality. This crossing is irreversible: recovering the off-diagonal elements requires accessing all environmental correlations, which Landauer's principle [?] renders thermodynamically prohibited at any finite temperature.

4.2 Information Produced at the Boundary

Each decoherence event produces classical information: the specification of which pointer state $|s_i\rangle$ was selected. For a system decohering into N distinguishable pointer states:

$$I = \log_2 N \quad \text{bits}. \tag{4}$$

Landauer's principle establishes that producing (equivalently, irreversibly encoding) one bit of classical information requires a minimum energy dissipation of:

$$\Delta E_{\text{Landauer}} = k_B T \ln 2 \tag{5}$$

at temperature T. This bound has been experimentally verified [??]. The irreversibility is not practical but thermodynamic: undoing the decoherence would require erasing the produced information, dissipating energy into the environment and increasing its entropy. The second law forbids the reverse.

4.3 The Nearest Encoding Surface

IAM's Law specifies that the cost is paid to the *nearest available encoding surface* at its local temperature T_H. In the local regime, for gravitationally collapsing systems, the nearest encoding surface is the Bekenstein-Hawking horizon of the forming black hole, at Hawking temperature $T_{\text{BH}} = \hbar c^3/(8\pi G M k_B)$. In the cosmological regime, the encoding surface is the Gibbons-Hawking cosmic horizon at temperature $T_H = \hbar H/(2\pi k_B)$ [?]. Both are thermal surfaces. The same law governs both. The scale determines which surface receives the cost: concentrated local collapses write first to the nearest black hole horizon; distributed large-scale structure formation writes to the growing cosmic horizon as it expands and cools. The transition between these two regimes is tracked quantitatively by the activation function $\mathcal{E}(a)$ derived in Section 6.

4.4 The Epoch-Dependent Cost

At the cosmological horizon, the cost per bit is epoch-dependent:

$$\Delta E_{\rm act}(a) = k_B T_H(a) \ln 2 = \frac{\hbar H(a)}{2\pi} \ln 2. \tag{6}$$

When H is large (early universe), bits are expensive. When H is small (late universe), bits are cheap. The total cost paid over cosmic history depends on both the rate of information production and the varying cost per bit — this is what gives rise to the activation function $\mathcal{E}(a)$, derived in Section 6.

5 Local Expression: The Complete Virial Theorem

5.1 Setup

Consider a system of total mass M transitioning from an unbound state to a stable bound equilibrium under a $1/r$ gravitational potential. Natural reference state: unbound at rest at infinity, total energy $E_i = 0$. The bound equilibrium state has total energy:

$$E_f = \langle K \rangle + \langle V \rangle. \tag{7}$$

No assumption about the ratio $\langle K \rangle / |\langle V \rangle|$ is made at this stage.

5.2 Step 1 — First Law: Heat Released at the Boundary

The transition from unbound to bound is irreversible. By the first law:

$$Q = E_i - E_f = -E_f = -(\langle K \rangle + \langle V \rangle). \tag{8}$$

This is exact. $Q > 0$ requires $E_f < 0$, satisfied for any stable bound state.

5.3 Step 2 — Second Law: Entropy Produced

The transition is irreversible. By the second law:

$$\Delta S = \frac{Q}{T_{\rm vir}}, \tag{9}$$

where $T_{\rm vir} = m\sigma^2/3k_B$ is the virial temperature of the bound system, with m the characteristic particle mass and σ the velocity dispersion.

5.4 Step 3 — Landauer's Principle: The Actualization Cost

By IAM's Law, the minimum thermodynamic cost of this irreversible transition — the Landauer cost of the information produced — is:

$$E_{\rm Landauer} = T_{\rm vir}\, \Delta S. \tag{10}$$

Substituting Eq. (9):

$$E_{\rm Landauer} = T_{\rm vir} \times \frac{Q}{T_{\rm vir}} = Q. \tag{11}$$

The virial temperature cancels exactly. The actualization cost equals the heat released at the virialization boundary *independent of temperature and independent of scale*. This is the scale-invariance of IAM's Law: the same thermodynamic identity holds from hydrogen atoms to galaxy clusters because the temperature cancels at every scale.

5.5 Step 4 — Euler's Theorem: The Virial Theorem Derived

Steps 1–3 establish:

$$E_{\text{Landauer}} = Q = -E_f = -(\langle K \rangle + \langle V \rangle). \tag{12}$$

To complete the identification of $\langle K \rangle$, we invoke the equilibrium condition of the $1/r$ potential. For any potential $V \propto r^n$, Euler's theorem for homogeneous functions of degree n requires at stable equilibrium $2\langle K \rangle + n\langle V \rangle = 0$. For $1/r$ potentials ($n = -1$):

$$2\langle K \rangle + \langle V \rangle = 0 \quad \Longrightarrow \quad -E_f = -(\langle K \rangle + \langle V \rangle) = \langle K \rangle. \tag{13}$$

This is the virial theorem — not assumed, but derived from Euler's theorem as the mathematical condition that defines stable equilibrium under a $1/r$ potential. Any system not satisfying Eq. (13) is either not yet at equilibrium or is unstable.

5.6 The Thermodynamic Completion

Substituting Eq. (13) into Eq. (12):

$$\boxed{E_{\text{Landauer}} = Q = T_{\text{vir}}\,\Delta S = -E_f = \langle K \rangle = \tfrac{1}{2}|\langle V \rangle|.} \tag{14}$$

The chain is exact at every step. The virial theorem $2\langle K \rangle + \langle V \rangle = 0$ is the *mechanical face*: the equilibrium ratio of the $1/r$ potential, known since Clausius [?]. The thermodynamic completion $\langle K \rangle = Q = T\,\Delta S = E_{\text{Landauer}}$ is the *thermodynamic face*: the physical identity of what K is, established here for the first time. These are not two separate results. They are the two faces of the same partition of $|\langle V \rangle|$ at the actualization boundary.

5.7 The Physical Meaning of K

$\langle K \rangle$ is not merely the average kinetic energy of bound particles. It is the thermodynamic cost of the system's transition from unbound potential to bound actuality — the heat Q released irreversibly at the virialization boundary, encoded on the nearest available surface at Landauer cost $k_B T \ln 2$ per bit. $\langle K \rangle$ is retained in the system as the kinetic energy of the bound particles. Q is encoded permanently as entropy at the boundary. Both are exactly half of $|\langle V \rangle|$ — one half paying the cost of existence, one half sustaining existence.

5.8 Why the $1/2$ is No Longer Unexplained

The ratio $\langle K \rangle / |\langle V \rangle| = 1/2$ arises because the $1/r$ potential is homogeneous of degree -1. Euler's theorem gives the mechanical result $2\langle K \rangle = |\langle V \rangle|$. IAM's Law gives the thermodynamic result: the system partitions $|\langle V \rangle|$ exactly in half because that is the unique partition at which the Landauer cost of the transition equals the kinetic energy of the resulting bound state. Any other partition would violate either the virial equilibrium condition or the Landauer cost of the transition. The $1/2$ is not a coincidence. It is a necessity.

5.9 Scale Invariance: 37 Orders of Magnitude

Because the virial temperature cancels in Step 3, the identity $\langle K \rangle = E_{\text{Landauer}}$ holds independently of temperature, mass, or size. Table 1 shows the verification across 37 orders of magnitude in physical scale.

Table 1: The virial 1/2 partition verified across 37 orders of magnitude. The temperature cancellation in Eq. (11) guarantees scale-invariance. The cosmological entry is the strongest quantitative test: 17 independent MCMC chains return $\beta_m = 0.1583 \pm 0.0033$ against the virial prediction 0.1575 at 0.2σ, with zero free parameters.

System	**Scale**	**Force**	**Verification**
Atoms (H–Xe, 20 elements)	10^{-10} m	Coulomb	Exact, machine precision [?]
Molecules (H_2–CO_2)	10^{-9} m	Coulomb	Exact at equilibrium geometry
Solar structure	10^{9} m	Gravity	$\sim$10% [?]
White dwarfs (Chandrasekhar)	10^{7} m	Gravity	Exact, $1.44\,M_\odot$ observed [?]
Galaxy clusters ($\sim$20)	10^{23} m	Gravity	$\sim$20% [?]
N-body virial efficiency	10^{22}–10^{25} m	Gravity	0.815 ± 0.025, 6 independent sims [?]
Cosmological β_m	10^{26} m	Gravity	0.1583 ± 0.0033, 0.2σ, 17 chains [?]

Scale range: 10^{-10} m to 10^{26} m — 36 orders of magnitude.

6 Global Expression: The Jacobson–Cai-Kim Chain Completed

6.1 What Einstein and Jacobson Were Missing

Einstein's field equations describe the geometry of actual spacetime. They are correct and complete as a description of the geometry that results from matter distribution. But GR has no account of the thermodynamic cost of actualization itself. It describes what happens *after* decoherence events produce classical matter. It is silent on what it costs the universe to produce that classical matter.

Jacobson [?] came the closest. He showed that the Einstein equations follow from the Clausius relation $\delta Q = T\,dS$ applied to local causal horizons. The derivation proceeds as follows. At any spacetime point p, a small spacelike 2-surface $\mathcal{P}$ defines a local Rindler horizon $\mathcal{H}$ with null tangent k^a and approximate boost Killing vector $\chi^a = -\kappa\lambda k^a$, where κ is the surface gravity. The Unruh temperature is $T = \hbar\kappa/(2\pi k_B)$ [?]. The heat flux across the horizon is:

$$\delta Q = -\kappa \int_{\mathcal{H}} \lambda\, T_{ab}\, k^a k^b\, d\lambda\, dA. \tag{15}$$

With entropy proportional to horizon area $S = \eta A$, the entropy variation via the Raychaudhuri equation gives:

$$\delta S = -\eta \int_{\mathcal{H}} \lambda\, R_{ab}\, k^a k^b\, d\lambda\, dA. \tag{16}$$

Setting $\delta Q = T\,\delta S$ and requiring this to hold for *all* null vectors k^a yields the Einstein field equation with $G = 1/(4\hbar\eta)$ [?]. The machine is: *specify an entropy functional, derive a gravity theory.*

The entropy functional was incomplete. It contained only $S_{\rm geo} = \eta A$. The Landauer cost of irreversible bulk information production — the cost paid each time a quantum system virializes and crosses from potential to actual — was not included. IAM's Law identifies this as the missing term.

Cai and Kim [?] extended Jacobson's argument to the FRW apparent horizon, recovering the Friedmann equations from the first law applied to the apparent horizon at $\tilde{r}_A = 1/H$ with area $A_H = 4\pi/H^2$, geometric entropy $S_{\text{geo}} = \pi/(GH^2)$, and Gibbons-Hawking temperature $T_H = \hbar H/(2\pi k_B)$ [?]. The same incompleteness applies: the entropy functional lacked the informational term.

IAM's Law identifies the missing term. The total entropy of the cosmic horizon has two components:

$$S_{\text{total}} = S_{\text{geo}} + S_{\text{info}}, \tag{17}$$

where $S_{\text{geo}} = A_H/4G$ is the Bekenstein-Hawking geometric entropy [??] that reproduces GR through the Jacobson mechanism, and S_{info} is the accumulated Landauer cost of every irreversible quantum-to-classical transition in the bulk since stable massive particles first existed.

6.2 Two Sources of Information

The total information production rate in the universe has two components:

Vacuum baseline (i_{vac}): Quantum vacuum fluctuations produce a constant, time-independent rate of information. This baseline is always present, independent of structure. It maps directly to the cosmological constant Λ through the Jacobson mechanism.

Structural complexity ($i_{\text{struct}}(a)$): Gravitational collapse, decoherence, and bound-state formation produce a time-dependent growing contribution. In the early universe, before significant structure exists, $i_{\text{struct}} \approx 0$. As galaxies form and the cosmic web develops, i_{struct} grows. As expansion dilutes matter and weakens gravitational collapse, the process self-regulates toward a finite asymptote. This term maps to $\beta_m \mathcal{E}(a)$ in the matter-sector perturbation equations.

6.3 The Cai–Kim Derivation and the IAM Modification

Setting $-dE = T_H\, dS_{\text{geo}}$ with the Misner-Sharp energy $E = \tilde{r}_A/2G$ and using $dS_{\text{geo}} = -(2\pi/GH^3)\, dH$ recovers the Friedmann equations [?]. Adding S_{info}:

$$-dE = T_H\, d\big(S_{\text{geo}} + S_{\text{info}}\big). \tag{18}$$

The geometric piece yields the standard Friedmann equations. The informational piece introduces an additional effective energy density. *A critical clarification before writing this explicitly: the term $\beta_m \mathcal{E}(a)$ is sourced by inhomogeneous gravitational structure formation. In an exact homogeneous FRW spacetime, structure formation vanishes and with it this term. Equation (19) below is the formal thermodynamic completion of the Jacobson–Cai-Kim entropy balance — it is not a universal homogeneous background modification applicable to all sectors. The observable consequences of this term enter exclusively through the timelike matter-sector perturbation equations, as derived in Section 7. The background FRW expansion and all photon-sector observables are unchanged.*

$$H^2 = \frac{8\pi G}{3}\rho_m + \frac{\Lambda}{3} + \beta_m \mathcal{E}(a)\, H_0^2. \tag{19}$$

This is not a modification of GR. It is GR's thermodynamic derivation completed with the full entropy budget.

6.4 Information Production Rate

The rate of irreversible information production from gravitational structure formation scales as:

$$\dot{I} \propto \rho_m(a) \cdot D(a)^{5/2} \cdot f(a) \cdot H(a), \tag{20}$$

where $\rho_m \propto a^{-3}$ is the matter density, $D(a)$ is the linear growth factor, $f = d\ln D/d\ln a$ is the growth rate, and $n = 5/2$ is the critical exponent derived in Section 8. The effective rate of informational entropy increase on the horizon is:

$$\frac{dS_{\text{info}}}{dt} = \frac{\dot{I}}{T_H \cdot A_H}, \tag{21}$$

where $1/T_H = 2\pi k_B/(\hbar H)$ reflects the Landauer encoding cost per bit at the horizon temperature, and $1/A_H$ converts total information to surface density consistent with the holographic bound [**??**].

6.5 The Activation Function Derived

The decoherence constraint requires the informational energy density to satisfy:

$$\dot{\rho}_{\text{info}} = \rho_{\text{info}} \cdot \frac{H}{a}. \tag{22}$$

This separable ODE integrates directly:

$$\rho_{\text{info}}(a) \propto \exp\left(\int_0^a \frac{da'}{a'^2}\right) = \exp\left(1 - \frac{1}{a}\right) \equiv \mathcal{E}(a). \tag{23}$$

Normalization $\mathcal{E}(1) = 1$ is set by definition. Therefore:

$$\boxed{\mathcal{E}(a) = \exp\left(1 - \frac{1}{a}\right).} \tag{24}$$

The properties of $\mathcal{E}(a)$ follow directly from IAM's Law:

- $\mathcal{E}(a) \to 0$ as $a \to 0$: no actualization cost before stable massive particles exist.
- $\mathcal{E}(1) = 1$: normalization today.
- $d\mathcal{E}/da > 0$ always: each decoherence event is irreversible. The ledger only grows.
- $\mathcal{E}(\infty) = e \approx 2.718$: the accumulated cost asymptotes without arriving. The universe matures rather than tears apart.
- $\mathcal{E}(z) = \exp(-z)$: pure exponential decay with redshift — the simplest possible form for a quantity growing from zero at early times to a finite value today.

The $1/a$ in the exponent has a precise physical meaning: $\mathcal{E}(a)$ is the exponential of the cumulative *information surface density* on the cosmic horizon. What drives expansion is not how much total information the universe has produced, but how densely that information is packed on the horizon surface.

6.6 The Variational Formulation

The thermodynamic result admits a variational formulation, connecting IAM's Law to standard field-theoretic language. Define the scalar field [?]:

$$\varphi(t) \equiv \ln \mathcal{E}(a) = 1 - \frac{1}{a}, \tag{25}$$

which satisfies the decoherence constraint $\dot{\varphi} = H/a$. This is not a Klein-Gordon equation. φ does not propagate independently; it tracks the cumulative decoherence history. The informational action is:

$$S_{\text{info}} = \int \left[-\frac{3H_0^2}{8\pi G} \beta_m e^{\varphi} + \lambda \left(\dot{\varphi} - \frac{H}{a} \right) \right] \sqrt{-g}\, d^4x, \tag{26}$$

where λ is a Lagrange multiplier enforcing the decoherence constraint. Variation with respect to g_{ab} recovers Eq. (19) exactly. The equation of state of the informational energy density follows from the continuity equation $\dot{\rho} + 3H(\rho + P) = 0$:

$$w_{\text{info}}(a) = -1 - \frac{1}{3a}. \tag{27}$$

This is a mildly phantom equation of state ($w < -1$) at all finite a, approaching -1 as $a \to \infty$. The phantom behavior reflects the continuous conversion of gravitational potential energy into informational entropy — not a violation of the null energy condition, because φ is a thermodynamic variable constrained by geometry, not a propagating field. $w_{\text{info}}(a)$ is a formal thermodynamic parametrization of the entropy completion; it is not a homogeneous dark-energy fluid implemented in the Boltzmann equations. The operational implementation of IAM in observable cosmology is through the effective matter-sector friction history described in Section 7, not through a literal new background source term.

Energy-momentum is conserved identically: $\nabla_\mu T^{\mu\nu}_{\text{total}} = 0$. No energy is created or destroyed. The formal thermodynamic representation Eq. (19) is consistent with the contracted Bianchi identity and the Einstein tensor; this consistency confirms the entropy-balance derivation is self-coherent, not that a new universal homogeneous fluid has been added to the Friedmann equations.

6.7 The Virial Theorem as Balance Sheet

IAM's Law tells you that actualization events carry a Landauer cost. The virial theorem tells you exactly how much energy crosses the actualization boundary at each virialization event. Together they give the cosmological coupling constant with zero free parameters.

For a self-gravitating system in virial equilibrium, half the gravitational binding energy goes into the geometric channel (spacetime curvature, standard GR) and half goes into the informational channel (the actualization cost, paid to the encoding surface). The informational energy density at $a = 1$ is therefore:

$$\rho_{\text{info}}(a{=}1) = \frac{1}{2}\rho_m = \frac{\Omega_m}{2}\rho_{\text{crit}}. \tag{28}$$

Since $\rho_{\text{info}}(1) = \beta_m \mathcal{E}(1)\,\rho_{\text{crit}} = \beta_m \rho_{\text{crit}}$:

$$\boxed{\beta_m = \frac{\Omega_m}{2}.} \tag{29}$$

This is derived, not fitted. The 1/2 partition of the virial theorem IS the coupling constant of IAM's Law at cosmological scales. With $\Omega_m = 0.315$ [?], the prediction is $\beta_m = 0.1575$.

6.8 The Dual-Sector Split

IAM's Law applies exclusively to matter on timelike worldlines.

Definition 1 (Decoherence eligibility)**.** *A degree of freedom undergoes gravitational decoherence if and only if it propagates on a timelike worldline ($d\tau > 0$).*

Decoherence is a dynamical process requiring a finite proper time interval $\Delta\tau > 0$. Photons propagate on null geodesics: $ds^2 = 0$, $\Delta\tau = 0$. A photon does not accumulate proper time. It does not virialize. It pays no actualization cost. This is a geometric consequence of spacetime structure, not an assumption.

Proposition 1 (Dual-sector coupling)**.** *For matter (timelike worldlines): $S_{\rm info} > 0$, yielding:*

$$\mu(a) = \frac{H^2_{\Lambda\rm CDM}(a)}{H^2_{\Lambda\rm CDM}(a) + \beta_m\,\mathcal{E}(a)} < 1. \tag{30}$$

For photons (null worldlines): $S_{\rm info} = 0$, yielding:

$$\Sigma(a) = 1 \quad (\textit{exact, geometric necessity}). \tag{31}$$

At $a = 1$: $\mu_0 \equiv \mu(z{=}0) - 1 = -\beta_m/(1+\beta_m) = -0.136$, with zero free parameters beyond ΛCDM.

The $\beta_m\,\mathcal{E}(a)$ term does not modify the background Friedmann equation. In an exact homogeneous FRW spacetime the Weyl tensor vanishes ($C_{\mu\nu\rho\sigma} = 0$), and any entropy functional sourced by inhomogeneous structure formation must vanish in this limit. The IAM modification operates exclusively at the perturbation level, entering through the effective Hubble friction for matter perturbations while leaving the background and photon sector unchanged.

7 Perturbation Theory: Deriving $\mu(a) < 1$ and $\Sigma(a) = 1$

The entropy completion of Section 6 is a formal thermodynamic statement. This section shows precisely how it enters observable cosmology: not as a universal homogeneous background modification, but exclusively through the timelike matter-sector perturbation equations. The background FRW expansion rate, photon geodesics, CMB power spectra, BAO positions, and lensing are all unchanged. The sole observable effect is suppressed matter growth via enhanced Hubble friction in the matter-sector growth equation.

7.1 The Informational Field Does Not Fluctuate

The scalar field $\varphi = 1 - 1/a$ is defined through the background apparent horizon radius $\tilde{r}_A = 1/H(t)$ and therefore depends only on the homogeneous expansion rate $H(t)$. In linear scalar perturbation theory, local overdensities $\delta\rho$ source metric potentials Ψ and Φ, but they do not shift the background-defined apparent horizon at first order. Therefore:

$$\delta\varphi = 0 \tag{32}$$

exactly, at all orders in linear perturbation theory. This is analogous to the cosmological constant, which satisfies $\delta\Lambda = 0$: both are properties of the spacetime geometry, not of the local matter content. Formally, perturbing the constrained action (Eq. (26)): variation with respect to λ gives the perturbed constraint $\delta\dot{\varphi} = 0$, and combined with the initial condition $\delta\varphi(a_i) = 0$, the field remains unperturbed at all times.

7.2 Standard GR Perturbations with Modified Matter-Sector Friction

With $\delta\varphi = 0$, the cosmological perturbation equations are identical to standard GR. The sole effect of the entropy completion is to introduce an additional contribution to the effective matter-sector expansion rate, denoted $H_{\rm eff,m}$, which enters only the matter growth equation as enhanced Hubble friction. The background FRW expansion rate is unchanged. In conformal Newtonian gauge ($ds^2 = a^2[-(1+2\Psi)d\tau^2 + (1-2\Phi)d\mathbf{x}^2]$):

Poisson equation:

$$k^2\Psi = -4\pi G\, a^2\, \rho_m\, \delta_m. \tag{33}$$

No anisotropic stress:

$$\Psi = \Phi \tag{34}$$

since $\delta\varphi = 0$ introduces no anisotropic stress from the informational sector.

Matter growth equation:

$$\ddot{\delta}_m + 2H_{\rm eff,m}\,\dot{\delta}_m - 4\pi G\,\rho_m\,\delta_m = 0, \tag{35}$$

where $H^2_{\rm eff,m} \equiv H^2_{\Lambda\rm CDM} + \beta_m\,\mathcal{E}(a)\,H_0^2$ is the effective matter-sector friction history derived from the entropy completion. These are standard GR perturbation equations. The only modification from ΛCDM is the enhanced Hubble friction term $2H_{\rm eff,m}\,\dot{\delta}_m$, which suppresses matter growth. This is the mechanism of $\mu(a) < 1$.

7.3 $\mu(a)$ and $\Sigma(a)$ as Ratios of Hubble Rates

Mapping to the μ–Σ framework, the effective gravitational coupling for matter perturbations is the ratio of Hubble rates squared:

$$\boxed{\mu(a) = \frac{E^2_{\Lambda\rm CDM}(a)}{E^2_{\rm eff,m}(a)} = \frac{H^2_{\Lambda\rm CDM}(a)}{H^2_{\Lambda\rm CDM}(a) + \beta_m\,\mathcal{E}(a)\,H_0^2} < 1.} \tag{36}$$

The suppression $\mu < 1$ arises because the enhanced Hubble friction in IAM reduces the effective matter density parameter: $\Omega_m^{\rm IAM}(a) < \Omega_m^{\Lambda\rm CDM}(a)$, so matter clusters less effectively than in ΛCDM. Numerical values at key redshifts: $\mu(z{=}0) = 0.864$, $\mu(z{=}0.5) = 0.948$, $\mu(z{=}1) = 0.982$, with $\mu \to 1$ at high redshift as $\mathcal{E}(a) \to 0$.

Since both potentials satisfy the unmodified Poisson equation (Eq. (33)) with no anisotropic stress (Eq. (34)), photon geodesics are determined by $(\Psi + \Phi)/2 = \Psi$ without modification. Therefore:

$$\Sigma(a) = 1 \quad \text{(exact)}. \tag{37}$$

The photon exemption at the perturbation level is a *consequence* of $\delta\varphi = 0$ — that φ is a horizon quantity, not a local field — not an additional assumption. CMB power spectra, CMB lensing, BAO angular positions, and supernova luminosity distances are unchanged. IAM and ΛCDM produce CMB TT spectra differing by less than 0.13% at $\ell > 30$ [?].

7.4 Uniqueness in the μ–Σ Parameter Space

The combination $\mu(a) < 1$ with $\Sigma(a) = 1$ is unique among known theoretical models. Most modified gravity theories predict $\mu > 1$ (enhanced growth) or correlated deviations in both μ and Σ:

- $f(R)$ gravity: $\mu > 1$, $\Sigma > 1$
- DGP braneworld: $\mu > 1$, $\Sigma \approx 1$
- General Horndeski: correlated deviations in both μ and Σ
- IAM: $\mu < 1$, $\Sigma = 1$ exactly

The IAM signature — weakened matter growth with unmodified lensing — is characteristic of a model that modifies matter-sector dynamics through Hubble friction without introducing anisotropic stress or touching photon geodesics. No other known framework produces this combination from first principles.

7.5 Numerical Validation

Level 1 validation uses MGCAMB v1.5.2 + Cobaya in the μ–Σ parametrization (12 chains, 4 dataset combinations: Planck only, Planck + RSD, Planck + BAO, Planck + Pantheon+). Level 2 validation uses direct Fortran modification of CAMB v1.5.8 + Cobaya implementing the IAM growth equation (Eq. (35)) via the matter-sector expansion rate `adotoa_matter` in the Boltzmann solver (5 chains). Two Level 2b exploratory runs applying the modification at the background Friedmann level produced $H_0 \approx 61.5$ km s^{-1} Mpc^{-1}, confirming that IAM operates exclusively at the perturbation level. All 17 chains converged to $R - 1 < 0.01$ with zero discarded runs [?].

8 The Bridge: $n = 5/2$ from Local to Global

The claim that local and global are the same law requires a quantitative bridge. That bridge is the critical exponent $n = 5/2$, derived independently from two directions.

8.1 Top-Down: From Horizon Thermodynamic Consistency

The top-down derivation requires the cumulative informational entropy to reproduce $\mathcal{E}(a) = \exp(1 - 1/a)$. Under matter-domination scalings ($H \propto a^{-3/2}$, $A_H \propto a^3$, $T_H \propto a^{-3/2}$, $D \propto a$), the integrand of Eq. (21) per unit $\ln a$ gives:

$$\frac{dS_{\rm info}}{d\ln a} \propto \frac{\rho_m \cdot D^n \cdot f}{T_H \cdot A_H} \propto a^{n-9/2}. \tag{38}$$

Converting to da and integrating:

$$S_{\rm info}(a) \propto \frac{a^{n-9/2}}{n - 9/2}. \tag{39}$$

For $S_{\rm info}(a) \propto -1/a + \text{const}$ (the form required by $\mathcal{E}(a) = \exp(1 - 1/a)$), we need $a^{n-9/2} = a^{-1}$:

$$n - \frac{9}{2} = -1 \quad \Longrightarrow \quad \boxed{n = \frac{5}{2}.} \tag{40}$$

8.2 Bottom-Up: From Structure Formation Physics

The bottom-up derivation integrates IAM's Law over the Press-Schechter halo mass function [?]. The rate of information production per virialization event is set by the Landauer cost at the virialization boundary (local IAM's Law). Integrating over the halo population with peak-height scaling $\nu \propto D(a)^{-1/2}$ near the characteristic mass M_*, the dominant contribution to the information production rate scales as $D^{5/2}$, giving $n = 5/2$ from quantum decoherence physics directly.

The Sheth-Tormen mass function [?] at $\sigma_* = 1.2$ (galaxy-scale halos where the bulk of cosmic information production occurs) recovers $\mathcal{E}(a) \propto \exp(0.925 - 1.009/a)$ with Pearson correlation $r = 0.992$ — the $1/a$ exponent to within 1% without free parameters. Four independent N-body studies spanning $z = 0$–30 [????] measure $n_{\text{eff}} = 3.33 \pm 0.33$, consistent with the IAM predicted range $n \approx 3.0$–3.5 under full ΛCDM evolution beyond the matter-domination approximation.

8.3 Convergence: One Law, Two Directions

Both directions yield $n = 5/2$. The top-down derivation requires it from horizon thermodynamic consistency. The bottom-up derivation derives it from the physics of quantum decoherence integrated over the halo population. Their agreement is a nontrivial consistency result: the same thermodynamic cost per bit, applied at the local decoherence boundary and integrated across the halo population, recovers the same critical exponent that horizon thermodynamics independently requires. This is strong support for one mechanism operating at two scales.

9 The Nine-Step Causal Chain

This section traces the causal sequence by which IAM's Law, applied to gravitational structure formation, produces the observable cosmological consequences derived in Sections 6 and 7. Each step follows from established physics; the references are to the standard results used at each stage.

1. **Gravitational interaction.** Matter interacts gravitationally, initiating collapse of overdense regions (GR [?]).
2. **Gravitational collapse.** Matter clusters into bound systems: halos, galaxies, stars (structure formation [?]).
3. **Quantum decoherence.** Gravitational interaction resolves quantum superpositions into definite classical states (Eq. (2), [?]).
4. **Irreversible information production.** Each decoherence event produces classical information ($I = \log_2 N$ bits, Eq. (4)) that cannot be recovered.
5. **Landauer thermodynamic cost.** Information production carries an irreducible energy cost of $k_B T_H \ln 2$ per bit at the horizon temperature (IAM's Law, Eq. (1)).
6. **Holographic encoding.** The produced information is bounded by and encoded on the cosmic horizon area [???], contributing S_{info} to the entropy functional.

7. **Informational pressure.** The modified first law $-dE = T_H\, d(S_{\rm geo} + S_{\rm info})$ introduces additional effective energy density $\beta_m\, \mathcal{E}(a) H_0^2$ in the matter-sector perturbation equations (Eq. (19)).

8. **Modified growth.** The matter-sector perturbation equations acquire enhanced Hubble friction, suppressing growth: $\mu(a) = H^2_{\Lambda{\rm CDM}}/H^2_{\rm IAM} < 1$.

9. **Feedback suppression.** Suppressed growth dilutes $\Omega_m(a)$, weakening gravitational collapse and reducing the amplitude of subsequent decoherence events.

Step 9 feeds back into Step 1 with reduced amplitude. The chain self-regulates, and $\mathcal{E}(a)$ asymptotes to e at late times rather than diverging. This is the mechanism by which the activation function acquires its finite upper bound.

Note that gravity appears at two points in the chain: as the initiating cause of decoherence (Step 1) and as the observable modified by the resulting informational pressure (Step 8). The $\beta_m = \Omega_m/2$ posterior in Section 12, consistent with the virial theorem prediction at 0.2σ, is the quantitative signature of this feedback structure.

10 Black Holes: The Local Saturation Limit of IAM's Law

The results of Sections 5–7 address the distributed regime of IAM's Law: structure formation producing decoherence events across many halos, encoding cumulatively on the growing cosmic horizon. This section addresses the concentrated local limit: what IAM's Law predicts when the local information production rate saturates the encoding capacity of the nearest bounding surface. The result is the standard black hole formation condition, recovered here from the informational saturation criterion rather than from gravitational collapse directly.

10.1 The Holographic Saturation Condition

The feedback in the nine-step chain becomes runaway when the local information production rate saturates the encoding capacity of the bounding surface. A black hole forms when the informational entropy from gravitational decoherence within radius R saturates the Bekenstein-Hawking bound on the bounding surface:

$$\frac{S_{\rm info}(R)}{A(R)} \geq \frac{1}{4\ell_P^2}. \tag{41}$$

Setting $S_{\rm info} = S_{\rm BH} = 4\pi G M^2/(\hbar c)$ at the Schwarzschild radius $R_s = 2GM/c^2$:

$$\frac{4\pi G M^2/(\hbar c)}{4\pi \cdot 4G^2M^2/c^4} = \frac{c^3}{4\hbar G} = \frac{1}{4\ell_P^2}. \tag{42}$$

The holographic bound is exactly saturated at the Schwarzschild radius. Eq. (41) is identically equivalent to the Thorne hoop conjecture [?]: the hoop conjecture is the holographic bound in geometric language; the saturation condition is the hoop conjecture in informational language.

10.2 The Two-Horizon Framework

Both the black hole horizon (temperature $T_{\rm BH} = \hbar c^3/(8\pi G M k_B)$) and the cosmic horizon (temperature $T_H = \hbar H/(2\pi k_B)$ [?]) are thermal encoding surfaces within the IAM framework. Setting $T_{\rm BH} = T_H$ yields the thermal equilibrium mass:

$$M_{\rm eq} = \frac{c^3}{4GH}. \tag{43}$$

At the present epoch, $M_{\rm eq} \approx 2.3 \times 10^{22} M_\odot$ — far exceeding the largest known black hole. All astrophysical black holes satisfy $T_{\rm BH} \gg T_H$: information flows unambiguously from hot local horizons to the cold cosmic horizon, at the thermally-limited rate $\Gamma = c^3/(1920\, GM \ln 2)$ bits s^{-1}. The cosmic horizon is the final repository; all locally-encoded information migrates to the global ledger.

10.3 Black Holes as Preferred Early-Universe Encoding Surfaces

The activation function $\mathcal{E}(a) = \exp(1 - 1/a)$ tracks the encoding transition quantitatively. At early times ($a \ll 1$), the cosmic horizon is small and the Gibbons-Hawking temperature is high, making cosmic horizon encoding inefficient. Information from early gravitational decoherence is encoded primarily on local black hole horizons. As the universe expands, large-scale structure formation produces distributed decoherence, the cosmic horizon grows and cools, eventually dominating as the encoding surface. The activation function is the quantitative record of this transition.

This suggests a natural thermodynamic context for the observed population of massive black holes at high redshift: in the early universe, before the cosmic horizon grows large enough to encode efficiently, black holes are the nearest available encoding surfaces. Within the IAM framework, this is a natural consequence of the saturation condition rather than an anomaly requiring separate explanation.

11 The Thermodynamic Arrow of Time and the Coincidence Problem

IAM's Law is irreversible by construction: each Landauer cost payment is permanent, and $S_{\rm info}$ can only grow. This section records two direct consequences of that irreversibility that are not independent predictions but structural properties of the activation function $\mathcal{E}(a)$ already derived.

The activation function $\mathcal{E}(a) = \exp(1 - 1/a)$ is monotonically increasing in a. Since decoherence events are irreversible and Landauer costs are permanent, $\mathcal{E}(a)$ cannot decrease. Expressed in redshift:

$$\mathcal{E}(z) = \exp(-z). \tag{44}$$

The thermodynamic arrow of time follows directly: the informational content of the universe increases monotonically with cosmic time. At recombination ($a \approx 10^{-3}$), $\mathcal{E}(a) \approx 0$; at the present epoch, $\mathcal{E}(1) = 1$; at late times, $\mathcal{E}(a) \to e$. The arrow is not an additional assumption. It is a property of $\mathcal{E}(a)$ inherited from the irreversibility of decoherence encoded in IAM's Law.

The coincidence problem — why the informational energy density $\rho_{\text{info}} \propto \mathcal{E}(a)$ dominates at approximately the same epoch as structure formation peaks — has a direct causal explanation within IAM. The informational pressure term $\beta_m \mathcal{E}(a)$ is sourced by structure formation. It grows when structure formation is active and self-regulates as suppressed growth reduces the decoherence rate. The coincidence between dark energy domination and the peak of structure formation is therefore a consequence of IAM's Law rather than a free parameter: the same process drives both.

12 Observational Consistency

12.1 The Primary Test: $\beta_m = \Omega_m/2$

The primary quantitative test of IAM's Law is the cosmological coupling constant $\beta_m = \Omega_m/2 = 0.1575$, derived from the virial theorem with zero free parameters. Seventeen MCMC chains were run against the full Planck 2018 likelihood (TT, TE, EE + low-ℓ + lensing). All 17 converged to $R - 1 < 0.01$ with no discarded runs, no parameter adjustments, and no excluded redshift ranges [?]. The β_m posterior is consistent with the prediction of IAM's Law at cosmological scales: the virial theorem predicted 0.1575 from first principles; the chains returned 0.1583 ± 0.0033, a 0.2σ agreement.

Table 2: Summary of quantitative consistency checks. All results derived from IAM's Law with zero free parameters beyond ΛCDM. The cosmological constant and baryon asymmetry entries represent quantities derived by IAM's Law that are free parameters in all other frameworks [?].

Quantity	**Prediction**	**Result**
β_m (virial theorem)	0.1575	0.1583 ± 0.0033 (0.2σ)
σ_8	0.800	0.7998 ± 0.0058 (0.1σ)
$H_0^{(\text{photon})}$	$67.16\ \text{km s}^{-1}\,\text{Mpc}^{-1}$	67.16 ± 0.47 (0.37σ from Planck)
$H_0^{(\text{matter})}$	$72.26\ \text{km s}^{-1}\,\text{Mpc}^{-1}$	72.26 (0.75σ from SH0ES)
$\Delta\chi^2$ vs ΛCDM	≤ 0	+0.54 (best-fit)
$\Lambda/\rho_{\text{vac}}$ (CC)	1.141×10^{-123}	1.14×10^{-123} (0.07%, 0 free params)
η (baryon asymmetry)	6.115×10^{-10}	$(6.115 \pm 0.037) \times 10^{-10}$ (0.36%, BBN prior off)

All 18 chains converged. Zero discarded runs. Zero parameter adjustments.
Green rows: quantities derived by IAM's Law that are free parameters in all other frameworks.

12.2 Independent N-body Confirmation

Six independent N-body simulation campaigns [??????] spanning 25 years measure the mass-weighted virial efficiency $\langle \eta_{\text{vir}} \rangle = 0.815 \pm 0.025$, consistent with the IAM prediction of 0.81. The combined measurement $\Omega_m \times f_{\text{coll}} \times \langle \eta_{\text{vir}} \rangle = 0.315 \times 0.62 \times 0.815 = 0.159 \pm 0.010$ agrees with $\beta_m = \Omega_m/2 = 0.1575$ to 1.0% with no free parameters. These simulations were performed without any knowledge of the IAM framework; their agreement is an independent cross-check.

12.3 The Sector-Dependent Hubble Split

The sector-dependent Hubble split $H_0^{(\text{matter})}/H_0^{(\text{photon})} = \sqrt{1+\beta_m} = 1.073$ is a direct consequence of Proposition 1. The two values are not a tension. They are measurements of two distinct physical quantities separated by the accumulated actualization cost of 13.8 billion years of structure formation. Both emerge from a single posterior with zero free parameters beyond ΛCDM.

Gravitational wave standard sirens provide an independent test. Binary neutron star mergers occur in galaxies — matter-sector objects — so their host galaxy recession velocities probe the matter-sector Hubble flow. IAM predicts:

$$H_0^{\text{sirens}} = H_0^{\text{CMB}}\sqrt{1+\beta_m} = 67.4\sqrt{1.1575} = 72.51\ \mathrm{km\,s^{-1}\,Mpc^{-1}}. \tag{45}$$

The GW170817 standard siren measurement gives $H_0 = 75.5^{+5.3}_{-5.4}\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ [?], consistent with the IAM prediction at 0.5σ. This is an independent consistency check from a completely different observable: no CMB, no distance ladder, no BAO. The same $\beta_m = \Omega_m/2$ derived from the virial theorem is consistent with the gravitational wave Hubble constant to within 0.5σ without adjustment.

12.4 Precision Measurement of $\mu_0 = -0.136$

IAM's Law derives $\mu_0 = -0.136$ from first principles with zero free parameters. Table 3 shows the precision with which upcoming surveys will constrain this quantity. Current data are already consistent with the prediction at 0.6σ. Tighter constraints from Euclid and DESI Year 5 will either sharpen the agreement or reveal a discrepancy — both outcomes carry information about the validity of the entropy-completion identification at the heart of IAM's Law.

Table 3: Precision with which upcoming surveys will measure the IAM's Law prediction $\mu_0 = -0.136$. The precision improves by roughly an order of magnitude from current data to Euclid + DESI Year 5 combined. Current data are already consistent with the prediction at 0.6σ [?].

Survey	$\sigma(\mu_0)$	**Precision on** $\mu_0 = -0.136$
DESI DR1 (current) [?]	0.22	0.6σ measurement
DESI Year 5 (projected)	~ 0.08	1.7σ measurement
Euclid DR1 (projected) [?]	~ 0.04	3.4σ measurement
Euclid + DESI Year 5	~ 0.03	4.5σ measurement

12.5 The Cosmological Constant and Baryon Asymmetry as Derived Quantities

The results reported in this subsection are qualitatively distinct from those above. The preceding subsections document IAM's Law producing predictions *consistent with* observational data — a necessary but not sufficient condition for a law. This subsection documents IAM's Law *selecting* two numbers that are free parameters in every other framework in physics. This is a different category of evidence.

The cosmological constant. The observed ratio $\Lambda_{\rm obs}/\rho_{\rm vac} = 1.14 \times 10^{-123}$ has resisted derivation for over fifty years. Within IAM, this ratio is the fraction of Planck-area horizon bits actualized by baryonic decoherence over cosmic history. The baseline prediction [?]:

$$\frac{\Lambda_{\rm obs}}{\rho_{\rm vac}} = \frac{2}{\pi}\left(\frac{l_P}{l_H}\right)^2 \frac{\Omega_b}{\Omega_m} = 1.38 \times 10^{-123} \tag{46}$$

recovers the observed value within a factor of 1.2 with zero free parameters. The factor arises from the departure from exact de Sitter geometry. The de Sitter encoding surface (the cosmological horizon) has a Gibbons–Hawking temperature $T_{dS} = T_H\sqrt{\Omega_\Lambda}$, which differs from the effective Hubble writing temperature T_H by the ratio:

$$\frac{T_{dS}}{T_H} = \sqrt{\Omega_\Lambda} = \sqrt{0.6846} = 0.8274. \tag{47}$$

This is a geometric correction, not a free parameter. Applying it:

$$\left.\frac{\Lambda_{\rm obs}}{\rho_{\rm vac}}\right|_{\rm corrected} = \frac{2}{\pi}\left(\frac{l_P}{l_H}\right)^2 \sqrt{\Omega_\Lambda}\,\frac{\Omega_b}{\Omega_m} = 1.141 \times 10^{-123} \tag{48}$$

against the observed 1.14×10^{-123} — agreement to within 0.07%, zero free parameters beyond the standard cosmological model. The 122-order-of-magnitude discrepancy is closed by a calculation involving only the Planck length, the Hubble constant, and the standard cosmological density parameters.

The baryon asymmetry. The baryon-to-photon ratio $\eta \approx 6.14 \times 10^{-10}$ is independently measured by Big Bang nucleosynthesis and CMB acoustic peaks, but not derived from a deeper principle in any existing framework. IAM's Law connects η to Λ through the information writing constraint: the CC formula (Eq. 46) contains Ω_b explicitly. Inverting it — asking what Ω_b the observed Λ requires — constrains η through $\eta \approx 2.74 \times 10^{-8}\,\Omega_b h^2$.

A dedicated MCMC chain was run with the BBN prior on $\Omega_b h^2$ removed and replaced with a flat prior four times wider than standard ($[0.010, 0.040]$, compared to the standard $[0.020, 0.025]$). The chain configuration was otherwise identical to the IAM Level 1 Run A used in the primary cosmological test [?]: MGCAMB v1.5.2 + Cobaya, full Planck 2018 likelihood suite, $\mu_0 = -0.13495$ fixed. The chain has no access to nuclear physics or light element abundance data. The baryon density is determined solely by the CMB acoustic peak structure.

The chain converged with $R - 1 = 0.009273$ after 22,400 accepted steps. The posterior for $\Omega_b h^2$ yields $\Omega_b h^2 = 0.022319 \pm 0.000136$, corresponding to:

$$\eta_{\rm IAM} = (6.115 \pm 0.037) \times 10^{-10} \tag{49}$$

against the observed $\eta_{\rm obs} = 6.137 \times 10^{-10}$ [??] — agreement to within 0.36%, with no nuclear physics input. Thus the baryon density selected by the CMB likelihood can be compared directly against the value required by the IAM cosmological-constant constraint, without importing BBN as an external prior. The prediction was stated in advance of the result [?]. No post-hoc adjustment was made.

The connection. The cosmological constant problem and the baryon asymmetry problem are not two independent problems within IAM. They are two expressions of the same information writing constraint: IAM's Law requires a specific baryonic matter density to produce the observed Λ, and the Planck CMB likelihood, with no external baryon constraint, returns a density consistent with that requirement. The same $\sqrt{\Omega_\Lambda}$ geometric correction that closes the CC calculation to 0.07% also accounts for the offset between the baseline CC-implied η and the CMB-only baryon result, suggesting a common physical origin within the IAM framework.

These results are documented in **?** with timestamps confirming all predictions were established prior to any chain runs.

13 Assumptions

Any paper claiming to state a law must be explicit about what is assumed and what is derived. The following assumptions underlie IAM's Law and its consequences derived in this paper.

Assumption 1 — Gravity is thermodynamic (Jacobson hypothesis). The Einstein field equations are an equation of state derivable from the Clausius relation applied to causal horizons with entropy proportional to area. This hypothesis is supported by the Bekenstein-Hawking entropy, the Unruh effect, and the Cai-Kim derivation of the Friedmann equations from the same thermodynamic argument. It is the foundation of Jacobson's 1995 result and is not universally accepted.

Assumption 2 — Decoherence-produced information contributes to horizon entropy. The Landauer cost of every irreversible quantum-to-classical transition in the bulk is encoded on the nearest available encoding surface and contributes to the entropy functional. This is the single new physical identification introduced by IAM's Law. It is consistent with the holographic principle [**??**] and Landauer's principle [**?**], but has not been independently verified at the level of a UV-complete quantum gravity theory. This is the same gap that exists in all thermodynamic approaches to gravity.

Assumption 3 — The virial coupling $\beta_m = \Omega_m/2$ is exact. The virial theorem in its strict form applies to systems in stable bound equilibrium. Cosmological structure formation is non-equilibrium. The derivation in Section 6 applies the virial partition to the ensemble of virialized structures at the population level. The 0.3% agreement between the virial prediction and the N-body measurement (Section 12) and the 0.2σ MCMC confirmation are empirical justifications, not proofs of exactness.

What is derived, not assumed. The activation function $\mathcal{E}(a) = \exp(1 - 1/a)$ is derived from integrating the decoherence constraint directly (Section 6). The dual-sector split $\mu < 1$, $\Sigma = 1$ is derived from the geometric fact that photons propagate on null geodesics ($d\tau = 0$). The critical exponent $n = 5/2$ is derived from both directions independently (Section 8). The prediction $\mu_0 = -0.136$ follows from these inputs with zero free parameters. None of these are assumed.

Three distinct failure modes. The assumptions above imply three separable levels at which the framework could be falsified, and these levels do not rise or fall together.

Level 1 (core law): Assumption 2 could fail. Decoherence-produced information may not contribute to horizon entropy in the way identified here. Falsification at this level would require a UV-complete theory that explicitly excludes this channel, or a derivation showing the entropy identification is internally inconsistent.

Level 2 (cosmological implementation): The virial coupling Assumption 3 could fail at the population level. A discrepancy between the predicted $\beta_m = \Omega_m/2$ and future MCMC posteriors — or a systematic failure of the perturbation-level growth suppression to fit upcoming survey data — would falsify the cosmological model without necessarily falsifying the core law.

Level 3 (broader structural implications): The extensions in Sections 9–11 and the discussion of the weak force and black holes as encoding surfaces are consequences suggested by the framework but not yet verified at the same level as the core derivation or the cosmological implementation. Failure of these extensions would not constitute falsification of either Level 1 or Level 2.

14 Discussion

14.1 The Three-Step Derivation Chain

Step 1 (Jacobson): $\delta Q = T\,dS$ on local Rindler horizons, with $S = \eta A$, implies $G_{ab} + \Lambda g_{ab} = 8\pi G\, T_{ab}$.

Step 2 (Cai–Kim): $-dE = T\,dS_{\rm geo}$ on the FRW apparent horizon, with $S_{\rm geo} = A_H/(4G)$ and $T = H/(2\pi)$, implies $H^2 = (8\pi G/3)\rho + \Lambda/3$.

Step 3 (IAM): $-dE = T\,d(S_{\rm geo} + S_{\rm info})$, with $S_{\rm info}$ from gravitational decoherence, implies $H^2 = (8\pi G/3)\rho + \Lambda/3 + \beta_m\,\mathcal{E}(a)\,H_0^2$.

Step 3 is the only new physical input: gravitational decoherence irreversibly produces classical information, which is holographically encoded on the cosmic horizon, contributing $S_{\rm info}(a)$ that grows monotonically with cosmic time. Every other element is established physics.

14.2 What Einstein, Jacobson, and Clausius Had

Clausius (1870): The mechanical face of the virial theorem — $2\langle K\rangle + \langle V\rangle = 0$ — derived from Newton's second law. Correct and complete as a mechanical statement. Silent on the thermodynamic meaning of K.

Einstein (1915): The geometry of actual spacetime. Correct and complete as a description of curved geometry. Silent on the cost of producing classical matter.

Jacobson (1995): Gravity as thermodynamics. GR derived as an equation of state from $\delta Q = T\,dS$. The machine: *specify an entropy, derive a gravity theory*. Missing one entropy term.

This work: The missing entropy term is the accumulated Landauer cost of actualization. Adding it closes Jacobson's derivation. The thermodynamic face of the virial theorem is the same cost at the local scale. IAM's Law is the statement that unifies both.

14.3 The Four Fundamental Forces

Three of the four fundamental forces satisfy the virial 1/2 partition. Gravity: confirmed from stellar structure to the cosmological coupling constant at 0.2σ across 17 MCMC chains. Electromagnetism: confirmed at machine precision for 20 atoms and 10 molecules from published Hartree–Fock energies [?], with mean virial ratio $\eta = 1.0000000000$, standard deviation zero — an exact verification because the virial theorem must hold exactly for converged $1/r$ wavefunctions. The strong nuclear force: the Chandrasekhar white dwarf limit $M_{\rm Ch} = 1.44\,M_\odot$ is an exact observational consequence of virial equilibrium between electron degeneracy pressure and gravitational collapse, confirmed [?].

The weak nuclear force does not satisfy the partition. This is consistent with the physical content of IAM's Law, and the reason is structural rather than coincidental.

The weak force is excluded by the law's own logic for three independent reasons. First, it violates parity [**?**]: the laws governing weak interactions are not symmetric under spatial reflection. Second, it violates CP symmetry [**?**]: it treats matter and antimatter differently, producing the baryon asymmetry of approximately one excess matter particle per 10^9 matter-antimatter pairs that constitutes all observable matter. Third, and most directly: a force whose function is to drive irreversible transitions *between* stable bound states cannot simultaneously satisfy an equilibrium condition within them.

The weak force is not one of four equals in the IAM framework. It is the *precondition* for IAM's Law to operate. Without CP violation and parity violation, there is no preferred direction for decoherence, no arrow of time at the particle physics level, no activation function $\mathcal{E}(a)$, and no IAM sector split. The other three forces maintain virial equilibrium in the stable structures that decoherence produces. The weak force produces the irreversibility that makes decoherence possible in the first place.

Three forces maintain stable equilibrium structures; one force drives irreversible transitions between them. IAM's Law governs the boundary between potential and actual; the weak force is the crossing. The fact that exactly three of four forces satisfy the equilibrium condition, and the one exception is the force whose function requires it to not satisfy it, is structurally consistent with IAM's Law identifying the boundary between quantum potential and classical actuality as a thermodynamically costly one. Whether this consistency constitutes independent evidence for the law or merely a post-hoc structural observation is a question the authors leave to the reader.

14.4 Where IAM's Law Sits in the Landscape of Physics

Every fundamental law in physics describes one side of a boundary. Classical mechanics describes the post-classical regime — the deterministic evolution of definite states. General relativity describes the geometry of actual spacetime — the curved manifold that results from the distribution of classical matter and energy. Quantum mechanics describes the pre-classical regime — the evolution of superpositions before any definite outcome is selected. Thermodynamics describes the statistical consequences of irreversible processes in the classical regime. The Standard Model describes the fields and particles that constitute classical matter once it exists.

Each of these frameworks is extraordinarily successful within its tested domain. Each has a gap at the same location: the boundary between quantum potential and classical actuality. Classical mechanics does not explain why the macroscopic world is classical. General relativity does not explain what it costs to produce the classical matter whose energy curves spacetime. Quantum mechanics describes superpositions but does not provide a mechanism for why superpositions become definite outcomes. Thermodynamics states that entropy increases but does not derive this from the dynamics of the quantum-to-classical transition. The Standard Model describes particles but not the process by which quantum fields become particles.

IAM's Law is a statement about the boundary itself.

The boundary is the decoherence event: the irreversible transition from quantum superposition to definite classical outcome described by Eqs. (2)–(3). IAM's Law states that this transition carries a thermodynamic cost (Eq. (1)), that this cost is irreversible, and that it is encoded on the nearest available encoding surface. This single statement has the following consequences, each connecting to an open problem in an established framework:

Connection to classical mechanics: The macroscopic world is classical because the actualization cost has been paid. Newton's laws operate in the post-actualization regime. IAM's Law is the mechanism that produces the regime in which Newton's laws hold.

Connection to general relativity: Jacobson's 1995 derivation [?] showed that the Einstein equations follow from $\delta Q = T\,dS$ applied to local causal horizons. The entropy functional was incomplete. IAM's Law identifies the missing term: the accumulated Landauer cost of every irreversible quantum-to-classical transition in the bulk. Adding it closes Jacobson's derivation. The cosmological model derived here is the observable consequence.

Connection to thermodynamics: The second law states that entropy increases but does not derive this from the dynamics of the quantum-to-classical transition. IAM's Law provides the mechanism: entropy increases because every crossing from potential to actual pays an irreversible Landauer cost to the nearest encoding surface. The second law is a corollary. The activation function $\mathcal{E}(a) = \exp(1 - 1/a)$, monotonically increasing and bounded above, is the quantitative record of this irreversibility applied across cosmic time.

Connection to quantum mechanics: The measurement problem — how quantum superpositions become definite classical outcomes — is reframed by IAM's Law from a foundational mystery into a thermodynamic transaction. The transition occurs when a quantum system interacts irreversibly with its environment, producing classical information at Landauer cost $k_B T_H \ln 2$ per bit. The transition is not instantaneous collapse but the irreversible orthogonalization of environmental states described by Eq. (2). IAM's Law does not solve the measurement problem at the level of a UV-complete quantum gravity theory; it identifies the thermodynamic cost of the transition and shows that this cost, accumulated over cosmic history, has observable cosmological consequences consistent with current data at 0.2σ across 17 independent MCMC chains.

Connection to the cosmological constant: The vacuum baseline of information production — quantum vacuum fluctuations producing a constant, epoch-independent rate of classical information — maps directly to Λ through the Jacobson mechanism. The cosmological constant is the encoding cost of the vacuum baseline. The structural complexity term $\beta_m \mathcal{E}(a)$ is the encoding cost of gravitational structure formation above that baseline. IAM's Law places both terms in the same entropy budget, unified by the same cost per bit. The quantitative consequence is derived in Section 12.5: applying the Gibbons–Hawking temperature ratio $\sqrt{\Omega_\Lambda} = 0.827$ closes the cosmological constant to within 0.07% of the observed value with zero free parameters, and simultaneously selects the baryon asymmetry $\eta = (6.115 \pm 0.037) \times 10^{-10}$ from the CMB alone.

The common thread is that each of these connections arises from the same physical fact: every major framework in physics has a gap at the quantum-to-classical boundary, and IAM's Law is a statement about that boundary. This is not a claim that IAM's Law solves all open problems in physics. It is a claim that those open problems share a common location, that IAM's Law operates at that location, and that its observable consequences are consistent with data to the precision currently available. The derivation is complete within its stated thermodynamic domain. The UV-complete microscopic account of why decoherence events couple to horizon degrees of freedom at exactly $k_B T_H \ln 2$ per bit remains an open problem shared with all thermodynamic approaches to gravity, including Jacobson's 1995 derivation. IAM's Law is presented as a candidate law at this same level of closure: derivationally self-consistent, with explicit assumptions, and with a gap at the deepest level that no current framework has yet closed.

14.5 A Law, Not a Framework

The major theories of physics are frameworks: collections of mutually consistent postulates, equations, and structures that together constitute a theory. General relativity requires the equivalence principle, the metric tensor formalism, the geodesic equation, the Einstein field equation, and the Bianchi identities — assembled by Einstein into a consistent structure, but not all consequences of a single statement. The Standard Model requires a gauge group, a Higgs mechanism, a fermion content, and 19 independently measured parameters. Thermodynamics comprises four independently stated laws. Quantum mechanics requires the Schrödinger equation, the Born rule, the measurement postulate, and the tensor product structure for composite systems — multiple independent inputs.

IAM is structurally different. Every result in this paper is a consequence of one equation:

$$\Delta E_{\text{act}} = k_B T_H \ln 2 \quad \text{per bit,} \tag{1}$$

applied consistently at two scales using only established physics. No additional postulates are introduced. The derivation chain is:

- The thermodynamic completion of the virial theorem (Section 5) follows from IAM's Law applied at each virialization event, with the virial temperature canceling exactly.
- $\beta_m = \Omega_m/2$ (Eq. 29) follows from the virial 1/2 partition as the balance sheet of IAM's Law.
- $\mathcal{E}(a) = \exp(1 - 1/a)$ (Eq. 24) follows from integrating IAM's Law over cosmic time through the Cai–Kim first law.
- $\mu(a) < 1$ (Eq. 30) follows from adding S_{info} to the Jacobson–Cai-Kim entropy functional.
- $\Sigma(a) = 1$ (Eq. 31) follows from photons propagating on null geodesics and paying no actualization cost.
- The arrow of time follows from the cost being irreversible: $\mathcal{E}(a)$ is monotonically increasing because each payment is permanent.
- Black holes as preferred early-universe encoding surfaces (Section 10) follow from IAM's Law saturating the Bekenstein-Hawking bound locally before the cosmic horizon grows large enough to encode efficiently.
- The coincidence problem resolution (Section 11) follows from dark energy being driven by the same actualization cost as structure formation — they are the same process at the same epoch by construction.
- The sector-dependent Hubble split follows from the dual-sector separation that the cost produces between timelike and null worldlines.

Not one of these results required an additional free parameter or an additional physical assumption beyond IAM's Law plus established physics. This structural property — that one equation generates its consequences rather than requiring them to be assembled — is what the authors mean by calling IAM's Law a law rather than a framework.

The distinction is not a claim about final status; it is a description of the derivational architecture.

Every boxed result in this paper traces back to Eq. (1) through a chain of established physics. To the authors' knowledge, no additional free parameter or physical assumption beyond Assumption 2 (Section 13) is required. The derivations are presented in sufficient detail for this claim to be verified independently.

14.6 Zero Free Parameters

The entire IAM modification to ΛCDM is determined by established physics and the independently measured matter density:

- $\mathcal{E}(a) = \exp(1 - 1/a)$: derived from horizon thermodynamics.
- The $1/a$ exponent: derived from information surface density, confirmed to 1% by the Sheth-Tormen mass function.
- $\mathcal{E}(1) = 1$: set by definition.
- $\beta_m = \Omega_m/2$: derived from the virial theorem.

IAM is a zero-parameter extension of ΛCDM. The framework introduces no new fields, extra dimensions, or modifications to the gravitational action beyond the identification of decoherence-produced information as a contribution to horizon entropy.

IAM's Law has been applied to derive additional results across a range of physical domains, including the M-σ relation for supermassive black holes [?], the lensing-dynamics discrepancy in galaxy clusters, the phantom crossing signature in DESI $w_0 w_a$ fits as an artifact of fitting a dual-sector expansion history with a single-sector model, the cosmological constant to within 0.07% with zero free parameters (Section 12.5), and the baryon asymmetry $\eta = (6.115 \pm 0.037) \times 10^{-10}$ returned by the Planck CMB likelihood with the BBN prior removed — consistent with the observed 6.137×10^{-10} at 0.36% with no nuclear physics input (Section 12.5). These results, with derivations and timestamps, are archived at github.com/hmahaffeyges/IAM-Validation (Zenodo DOI: 10.5281/zenodo.18702042).

Acknowledgments

The thermodynamic framework of T. Jacobson [?] is the foundation upon which this work is built. His 1995 derivation identified gravity as thermodynamic and established the machine — specify an entropy functional, derive a gravity theory — that IAM's Law completes. A personal response from Professor P.J.E. Peebles identified the quantum-to-classical derivation as the foundational gap requiring closure before the cosmological framework could be trusted; that identification directly motivated the bridge from local decoherence physics to the global cosmological ledger. The decoherence framework of W.H. Zurek [?] provided the local account of the quantum-to-classical transition that the bridge extends to cosmological scales. The author thanks R. Clausius [?] and R. Landauer [?] for the tools that make this connection possible.

Data Availability

All MCMC chains, modified Fortran source, analysis scripts, and prediction figures are publicly archived at https://github.com/hmahaffeyges/IAM-Validation with canonical Zenodo DOI 10.5281/zenodo.18702042. Repository timestamps confirm all predictions were established prior to any observational comparison.

References